Adrian Garcia — Senior Full-Stack Engineer

Merced, CA | Open to Remote | Available Immediately

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Portfolio: https://portfolio-adiran-garcia.vercel.app

Senior Full-Stack Engineer with 8+ years building reliable, high-performance systems across fintech, healthcare, trading research, logistics, and data platforms. Specializes in Python/FastAPI, Node.js/TypeScript, React, distributed services, real-time data pipelines, and cloud-native architectures. Experienced in designing scalable APIs, event-driven backends, and end-to-end product features with a strong focus on performance, correctness, and observability. Integrates LLM/RAG components, vector search, and analytics workflows to deliver intelligent, production-ready systems that drive measurable impact across critical domains.

CORE SKILLS

Backend Frameworks & Microservices

- Python: FastAPI, Django/DRF, Flask, asyncio, Starlette, aiohttp, Celery, Pydantic, SQLAlchemy
- Node.js: Express.js, Fastify, NestJS, Hapi, Koa, tRPC, Apollo Server (GraphQL), Socket.IO
- Messaging & async: Celery, RabbitMQ, Kafka, Redis Streams, SQS/SNS, worker queues (BullMQ, Agenda), clustering, streaming APIs, Node event-loop profiling, high-throughput microservices optimized with async concurrency
- API styles: REST, gRPC, WebSockets/SSE; idempotency keys, retries/backoff, rate limiting, pagination, OpenAPI

Programming Languages

- Python, Go (Golang), TypeScript/JavaScript, Bash, SQL

Frontend Engineering

- React, Next.js, TypeScript, Hooks, Zustand/Redux, Vite
- Dashboards, real-time charts, WebSockets, data visualization (D3.js, Chart.js)

Data, Search & Storage

- PostgreSQL/TimescaleDB, MySQL, SQLite, Redis (cache/Streams), MongoDB
- Prisma, TypeORM, Sequelize, Knex; migrations, connection pooling, query performance tuning
- Vector DBs: Pinecone, FAISS, Milvus, Weaviate, Qdrant, pgvector
- Elasticsearch/OpenSearch
- Query optimization, indexing, connection pooling, caching, read/write segregation

Architecture & Reliability

- Microservices, event-driven systems, CQRS-lite, hexagonal architecture
- Concurrency patterns, backpressure, circuit breakers, token-bucket limiters
- API gateways & service mesh: Kong, Envoy, Traefik, Istio/Linkerd
- Observability: OpenTelemetry tracing, Prometheus/Grafana, Jaeger, ELK/Opensearch, structured logging

Cloud, DevOps & Infrastructure

- AWS: ECS/EKS, Lambda, RDS, S3, CloudFront, IAM, Step Functions, EventBridge
- Container & IaC: Docker, Kubernetes, Helm/Kustomize, Terraform
- CI/CD: GitHub Actions, GitLab CI, Jenkins; blue-green/canary deploys, artifact/versioning
- GPU/compute: AWS Batch, model-serving containers, TensorRT (light)

AI/ML & RAG Systems

- LangChain, LlamaIndex, LangGraph
- Guardrails: PII redaction, prompt safety, JSON-schema validation, cost/latency monitoring

Process & Collaboration

- Agile/Kanban, Jira/Confluence, code reviews, technical writing
- Testing: unit/integration/contract tests, mocks, load testing (k6/Locust)

PROFESSIONAL EXPERIENCE

Toptal ▪ San Francisco, CA — Senior *Backend and AI Engineer (Contract, Part-Time)*
Remote | Dec 2025 – Present

- Built and productionized a NER Extraction Service for Person/Organization/Location entity extraction from news content using a fine-tuned Transformer token-classification model, including evaluation-aligned post-processing for accurate multi-word entities.
- Delivered low-latency FastAPI inference API for both real-time and batch workloads, meeting specification requirements.
- Added a Redis-backed inference cache controlled via environment configuration to improve repeat-request performance and stabilize tail latency under load.
- Validated performance with Locust achieving ~5 ms median / 220 ms p99 with Redis enabled versus 38 ms median / 710 ms p99 without Redis (GPU run), with 0% failures.
- Containerized the service for both CPU and GPU execution and provided Docker Compose orchestration for Redis-enabled deployments.
- Implemented environment-driven runtime configuration (automatic device selection, model path configuration, optional Hugging Face Hub loading) to ensure consistent behavior across environments.
- Added unit and integration test coverage for core API and caching behavior (health checks, inference endpoints, Redis cache tests) to improve maintainability and release confidence.

Harvest LLC ▪ Wilmington, DE —*Backend & AI Engineer (Contract)*
Remote | Apr 2025 – Oct 2025

Core Services & APIs

- Designed and implemented Go and Python/FastAPI microservices for real-time query and ingestion workloads, featuring idempotent endpoints, OAuth2/JWT auth, RBAC, rate limits, retries/timeouts, and correlation-ID–based tracing.
- Delivered robust REST and gRPC APIs with strongly typed schemas and OpenAPI contracts; optimized pagination, backpressure, and connection pooling to increase throughput and API stability under load.
- Implemented Redis-backed caching layers (response, metadata) and concurrency tuning strategies that significantly reduced p95 latency during sustained traffic spikes.

Data Pipelines & Quality Engineering

- Developed production ETL/ELT pipelines for WDI, Eurostat, and IMF datasets, including incremental ingestion, normalization, checksums/deduplication, and automated schema-drift detection.
- Standardized storage using S3/Parquet with PostgreSQL/Supabase metadata; introduced lineage tracking and reproducible snapshotting for audits and data governance.
- Built validation suites and integrity checks that improved data accuracy and reduced pipeline failure rates while strengthening data freshness SLAs.

Platform, Operations & Reliability

- Containerized services with Docker and deployed to AWS ECS/EKS using ALB, autoscaling, health checks, and zero-downtime rollouts.
- Implemented secure CI/CD workflows with image scanning, unit/contract tests, staged promotions, and blue-green/canary deployments to reduce release incidents.
- Integrated OpenTelemetry for distributed tracing, metrics, and structured logs; built cost, latency, and SLO dashboards to enhance observability and operational decision-making.

AI Integration (Lightweight)

- Implemented internal RAG/semantic-search pipelines using LangChain + Pinecone with hybrid retrieval, reranking, and citation-aware responses.
- Added prompt-safety validation, guardrails, and fail-closed behaviors for low-confidence results; monitored token usage and compute cost to ensure efficient AI operations.

Impact

- Improved p95 latency, increased sustained throughput, stabilized ingestion pipelines, and reduced operational + LLM token costs through optimized caching, routing, and observability.
- Delivered a reliable and explainable semantic-search capability that expanded internal analytics functionality and boosted platform performance and reliability

Citadel LLC ▪ Chicago, IL, US— ***Software Engineer (Full-Stack, Part-Time)***

Remote | Dec 2023 – Mar 2025

Backend Architecture & API Development (Python FastAPI, Node.js, Express)

- Designed, built, and maintained core backend services using Python/FastAPI and Node.js/Express, delivering secure, high-performance APIs for financial and compliance applications.
- Implemented async REST services with JWT/OAuth2 authentication, rate limiting, request validation, and logging pipelines for reliability and traceability.
- Developed scalable backend workflows using Celery + RabbitMQ, with Redis for caching, session management, and high-throughput job coordination.

- Built modular service layers, reusable API contracts, and integration logic for internal data systems and external third-party providers.

Frontend Engineering (React, Next.js)

- Implemented responsive user interfaces, dashboards, and admin tools using React, integrating seamlessly with FastAPI/Node.js backend services.
- Built reusable UI components, state-management logic, and form workflows to support financial operations and document management features.
- Improved usability, performance, and load times by optimizing API consumption patterns, batching requests, and adopting client-side caching.

Full-Stack Feature Development & System Integration

- Delivered end-to-end features spanning backend APIs, frontend components, database schemas, and deployment workflows.
- Integrated payment, authentication, document, and internal analytics systems across multiple services using structured API definitions.
- Collaborated with product, design, and operations teams to refine requirements and ensure consistent user experience across the stack.

Cloud Infrastructure, CI/CD & DevOps

- Containerized full-stack applications using Docker and deployed them to Kubernetes clusters through Terraform-managed environments.
- Added structured logging, metrics dashboards, and health checks for proactive system monitoring and faster issue resolution.
- Maintained CI/CD pipelines (GitHub Actions, GitLab CI, Jenkins) with automated testing, image scanning, and staged deployments to improve release reliability.

Performance Optimization & Reliability

- Optimized database queries, API routes, caching strategies, and background jobs to improve p95 latency and system throughput.
- Enhanced backend reliability through structured error handling, retry policies, load balancing, and efficient use of queues/workers.
- Improved application performance and reduced operational overhead by tuning infrastructure, scaling worker counts, and refining deployment processes.

Convoy Inc ▪ Seattle, WA — *Senior AI & GenAI Engineer (Full-Time)*
Remote | Oct 2022 – Nov 2024

RAG Pipeline Architecture & Document Processing

- Architected an end-to-end document ingestion pipeline using Python and Apache Airflow to process legal PDFs and HTML content, applying spaCy-based text segmentation and generating semantic embeddings with Sentence Transformers (all-MiniLM-L6-v2), then indexing them in FAISS for high-throughput vector search.
- Engineered dense retrieval workflows by encoding user queries using the same embedding model, executing top-k similarity search in FAISS, and applying cross-encoder reranking to maximize context precision and retrieval quality.

- Designed advanced prompt-augmentation strategies in LangChain, dynamically constructing optimized LLM prompts by blending retrieved context with user queries while enforcing token-budget constraint.

LLM Integration & Context-Grounded Generation

- Integrated HuggingFace Transformer models (GPT-3, BERT, Electra) into retrieval pipelines to deliver context-grounded, legally compliant responses by combining parametric model knowledge with retrieved document insights.
- Built and deployed scalable RAG-driven Q&A microservices using FastAPI, encapsulating retrieval, prompt construction, and generative reasoning into low-latency, production-ready APIs.
- Implemented automated knowledge base refresh pipelines via AWS Lambda and S3 event triggers, allowing continuous ingestion, re-embedding, and FAISS index updates without service interruption.

Scalable AI Infrastructure & Deployment

- Designed and optimized GPU-accelerated inference environments using Docker, Kubernetes, and TensorRT, enabling high-throughput, low-latency execution for LLM and embedding workloads.
- Implemented distributed processing pipelines for large-scale embedding generation and vector indexing using Airflow, AWS Batch, and parallelized Python workers, significantly reducing end-to-end processing time Built secure, modular infrastructure-as-code (IaC) deployments using Terraform and AWS services (ECS, EKS, Lambda), ensuring reproducible, fault-tolerant AI environments that support continuous model updates.

Evaluation, Governance & Security

- Developed evaluation frameworks to track retrieval quality and LLM accuracy using recall, BLEU, ROUGE, and custom scoring metrics, with human-in-the-loop review processes ensuring correctness in legal workflows.
- Secured RAG infrastructure—vector databases, embedding pipelines, and inference endpoints—using OAuth2 authentication, encrypted storage, and strict access policies to protect sensitive legal documents.
- Established observability and monitoring for embedding drift, retrieval effectiveness, and generation consistency, enabling proactive model improvements and governance across the RAG ecosystem.

Zillow ▪ Settle, WA — ***Senior Software Engineer (Full-Stack, Full-Time)***
Remote | Jun 2019 – Oct 2022

Full-Stack Architecture, Web Applications & High-Performance Services

- Built scalable full-stack applications using Golang (Gin, Echo, gRPC, go-kit) and Python (FastAPI) on the backend, and modern JavaScript frameworks for responsive, high-performance UIs.
- Designed end-to-end features spanning backend APIs, data access layers, and frontend integrations, improving page load times and user workflows across multiple internal tools.
- Implemented real-time Web UI components powered by event-driven backends, improving data freshness and interactivity for analytics and operational dashboards.

Frontend Engineering, UI Integration & Client-Facing Systems

- Developed modular, reusable UI components and frontend logic using React/TypeScript, integrated with backend services through REST and gRPC endpoints.

- Improved frontend performance via caching strategies, API batching, and optimized state management.
- Built admin and internal dashboards supporting live data updates, system observability, and ML insights.

Event-Driven Systems, Data Pipelines & Search Infrastructure

- Created real-time ingestion and processing pipelines using Kafka, Golang consumers/producers, and Python ETL services.
- Integrated vector databases (Pinecone, Milvus, Weaviate) to power semantic search and recommendation features consumed directly by frontend applications.
- Designed optimized database schemas across PostgreSQL, Redis, and MongoDB to support both REST and UI-intensive queries.

API Engineering, System Integrations & Full-Stack Communication

- Built secure, well-structured REST and gRPC APIs enabling fast communication between client apps and distributed services.
- Implemented authentication flows, request validation, caching layers, and API gateway patterns supporting cross-frontend usage.
- Enhanced application reliability via structured logging (Zap, Logrus), Prometheus metrics, and distributed tracing visible in internal dashboards.

ML-Powered Features, Smart Automation & User-Facing Intelligence

- Integrated ML models into production systems, enabling real-time scoring, recommendations, anomaly detection, and ranking displayed in frontend tools.
- Built async workers and model-serving adapters connecting Python ML services to Go microservices and the frontend UI.
- Developed monitoring workflows to expose model metrics and performance in user dashboards

DevOps, CI/CD, Testing Automation & Cross-Team Engineering Leadership

- Built CI/CD pipelines with GitHub Actions + Docker, enabling rapid, stable deployments across frontend and backend services.
- Led architecture reviews, mentored full-stack and backend engineers, and improved testing culture across teams.
- Implemented blue-green deploys, canary rolls, automated test suites, and performance benchmarking for both API and UI layers.

Twills ▪ NY, NY, US— ***Full Stack Engineer (Part-Time)***
Remote | Nov 2017 – Apr 2019

Backend Engineering (Node.js & Python)

- Re-architected and scaled a high-throughput auto-parts e-commerce backend using Node.js/Express and Python services, delivering robust ordering, account management, and product-catalog APIs supporting 1,000+ daily users.
- Implemented enterprise-grade security for fintech operations, including JWT/OAuth2 authentication, RBAC, idempotent webhooks, audit logging, rate limiting, and strict request validation, ensuring regulatory compliance and safeguarding sensitive transactions.

- Developed reusable service adapters and integrations for payment processors, shipping carriers, and inventory systems, standardizing partner communication with OpenAPI contracts to maintain reliability and consistency.

API Development, Data Systems & Performance Optimization

- Designed and maintained high-availability B2B and internal APIs for real-time data exchange between back-office systems, customer-facing dashboards, and external vendor feeds, incorporating SLA monitoring and automated alerting.
- Optimized relational databases (MySQL/InnoDB) via advanced indexing, query plan tuning, connection pooling, and read/write segregation, reducing p95 query latency and overall system load.
- Built background workers and asynchronous pipelines (Celery, queues, task processors) to handle bulk imports, reconciliation, and email workflows, improving frontend responsiveness and operational throughput.

Cloud, DevOps & Reliability Engineering

- Led the migration from on-premise infrastructure to cloud-ready, containerized environments using Docker and VMware, implementing CI/CD pipelines for predictable, low-downtime deployments.
- Introduced centralized logging, monitoring, health checks, and automated incident alerting, enhancing observability and enabling proactive system maintenance.
- Established robust backup/restore processes, disaster recovery playbooks, automated testing, and environment parity standards, ensuring platform resilience and operational reliability in financial services contexts.

Frontend Engineering (React & Next.js)

- Modernized internal admin dashboards and catalog management tools with React and Next.js, improving workflows for search, pricing, and inventory operations.
- Delivered responsive, API-driven UIs aligned with WCAG accessibility standards, ensuring maintainable, performant interfaces for internal stakeholders.
- Collaborated closely with designers, backend engineers, and product managers to maintain seamless data flow, intuitive interactions, and consistent end-to-end system performance.

EDUCATION

- **UC Irvine —** PhD, Material Science **with Concentration in Computational Science / Data Science** · 2014-2019

  Research in simulation/data analysis; heavy use of **Python**, **C/C++**, **NumPy/Pandas**, **ML/Stats**, **HPC** (MPI/SLURM)

- **UC Irvine —** MS, MSE **(Scientific Computing & Analytics)** · 2012-2014

  Relevant CS topics: **scientific computing**, **numerical methods**, **data processing**, **visualization**

- **UC Merced —** BS, ME **(Programming & Modeling focus)** · 2008-2012

  Emphasis on **computational modeling** and **programming** (MATLAB/**Python**/**C++**)

PERSONANL PROJECTS

- **DataHaven.tech — AI Data Visualization Platform (MVP).**
  Designed and developed a full-stack analytics platform with a Python/FastAPI backend and a TypeScript-based frontend using React, D3.js, and Chart.js. Built ingestion and normalization pipelines to process large economic datasets from WDI, IMF, and Eurostat, ensuring clean schemas, efficient indexing, and responsive data queries. Implemented Node.js services for caching, API aggregation, and data transformation to support real-time insights and user-driven Q&A flows. Integrated secure backend features including JWT/OAuth2 authentication, idempotent API endpoints, and rate limiting for dependable production behavior. Delivered interactive dashboards with dynamic charts, filtering tools, and drill-down visualizations. Added monitoring, logging, and performance metrics to ensure stable, observable system behavior across the full stack.

- **Company Data Analytics — End-to-End Data Analytics & Predictive Modeling Pipeline.**
  Built an end-to-end data platform with a Python/FastAPI backend and lightweight Node.js/TypeScript utilities for scraping, transformation, and external API integrations. Developed workflows to collect, clean, and structure company data from Glassdoor and corporate websites, supporting downstream analytics and reporting. Orchestrated data pipelines with Airflow, including dependency management, retries, alerts, and automated recovery from pipeline failures. Exposed standardized REST APIs for embeddings, processed datasets, and predictive results, enabling easy integration with dashboards and internal tools. Generated automated analytical reports using Jupyter/Quarto with visualizations and clear summaries for decision-makers. Implemented data validation, schema enforcement, and quality checks to ensure accurate outputs and seamless use across frontend and backend components.

- **End-to-End Trading Platform — Systematic Trading Research & Real-Time Analytics.**
  Designed and developed a full-stack trading research platform supporting real-time market data ingestion, feature engineering, strategy backtesting, and risk analytics. Built an event-driven backend using Python (FastAPI) and Go microservices with WebSocket streaming, low-latency data pipelines, and TimescaleDB for high-frequency time-series storage. Implemented a modular strategy engine for simulation, PnL generation, Sharpe/drawdown metrics, and automated alerting. Delivered a fully interactive React/TypeScript frontend with rich dashboards for multi-asset performance, live market charts, and risk monitoring. Integrated Redis caching, idempotent APIs, authenticated endpoints, and containerized services for reliable production deployment across AWS.

LANGUAGE

- English: Native
- Spanish: Conversational